SMART COMPLAINT MANAGEMENT SYSTEM

SYSTEM VALIDATION REPORT

Functional workflow and integration validation

Project detail	Value
Group	8
Course	Software Engineering (UE24CS341A)
Institution	PES University
Version	1.0
Date	11 November 2026

PREPARED BY

Name	SRN	Role / Focus Area
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1. Objective

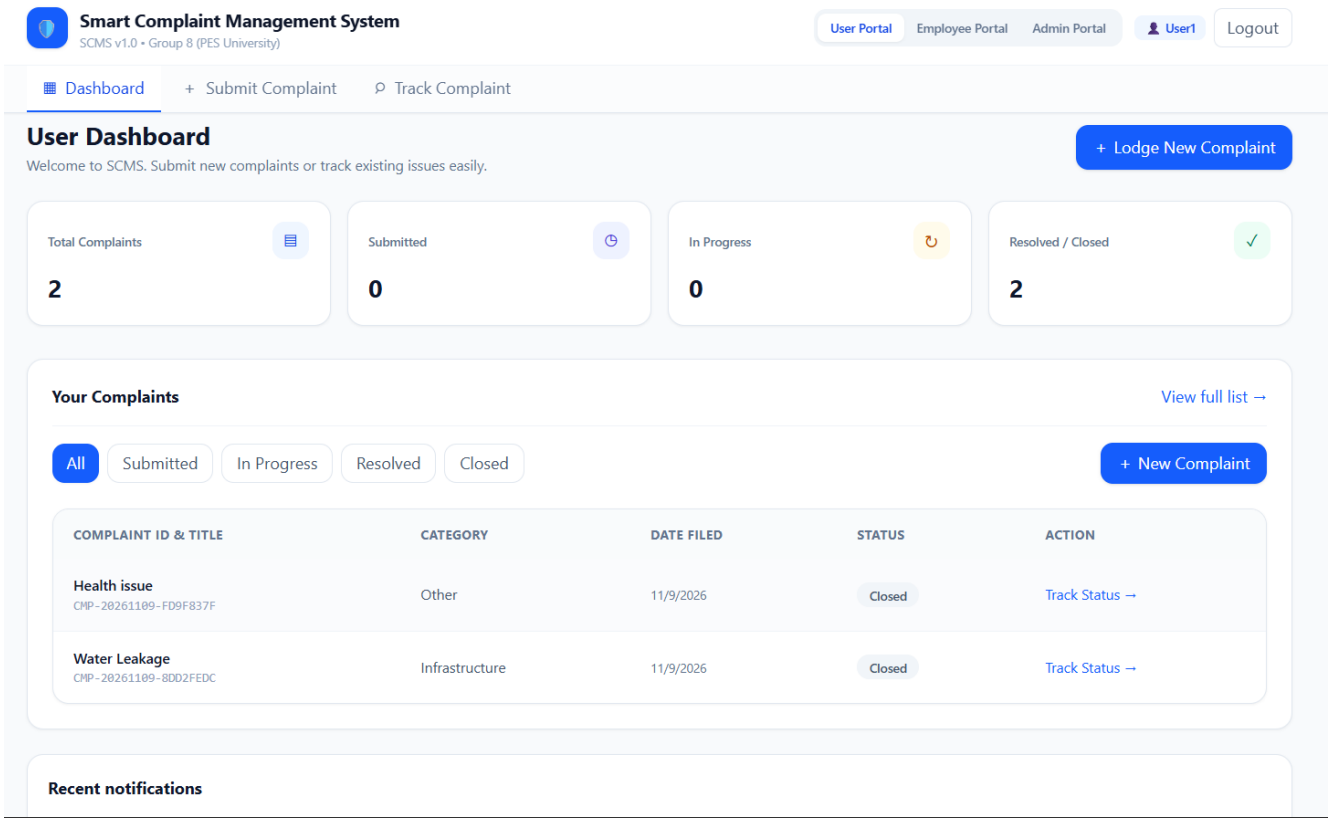
This report summarizes validation of SCMS as a centralized platform for users to submit and track complaints and for authorized personnel to process, resolve, and close them.

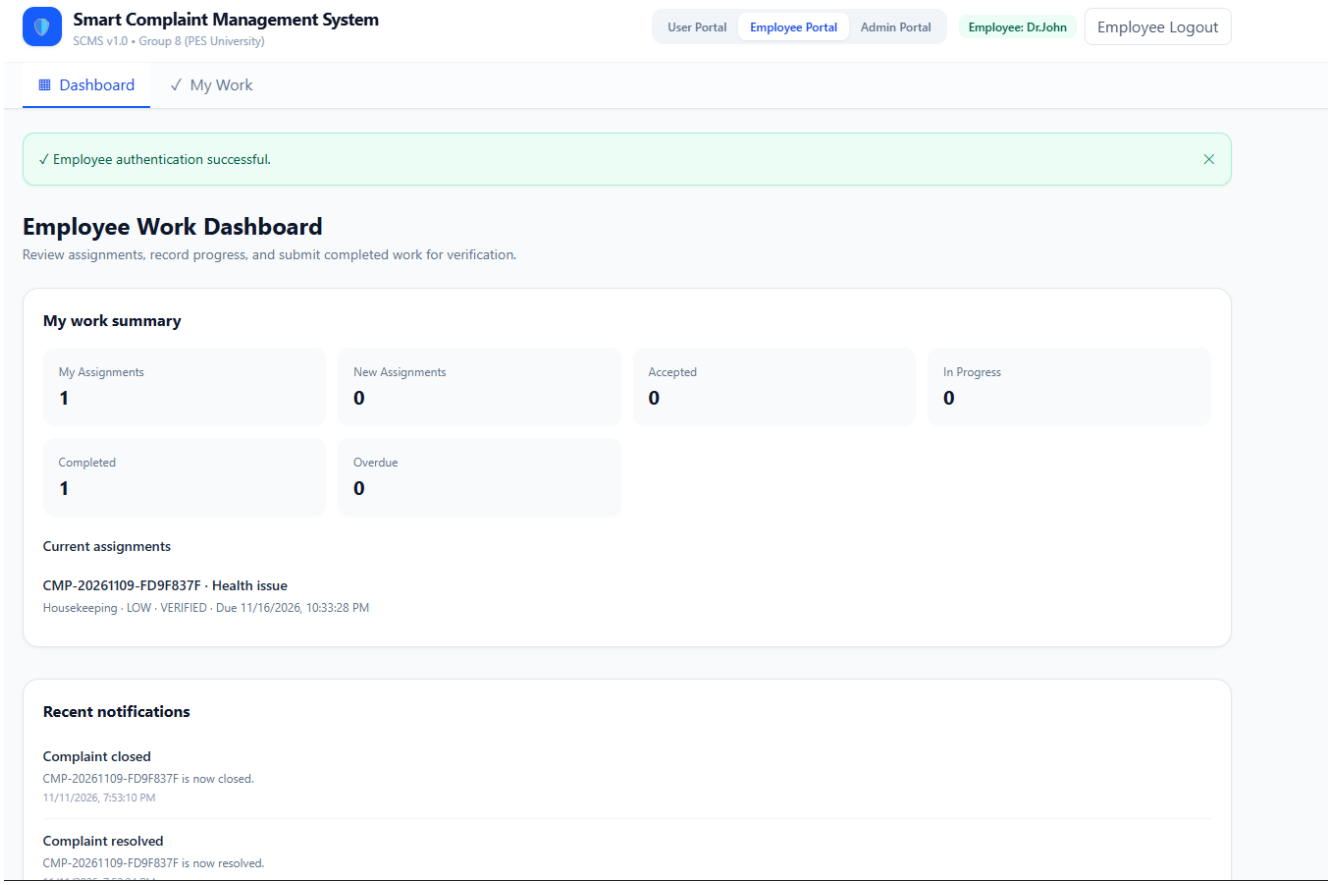
2. System Overview

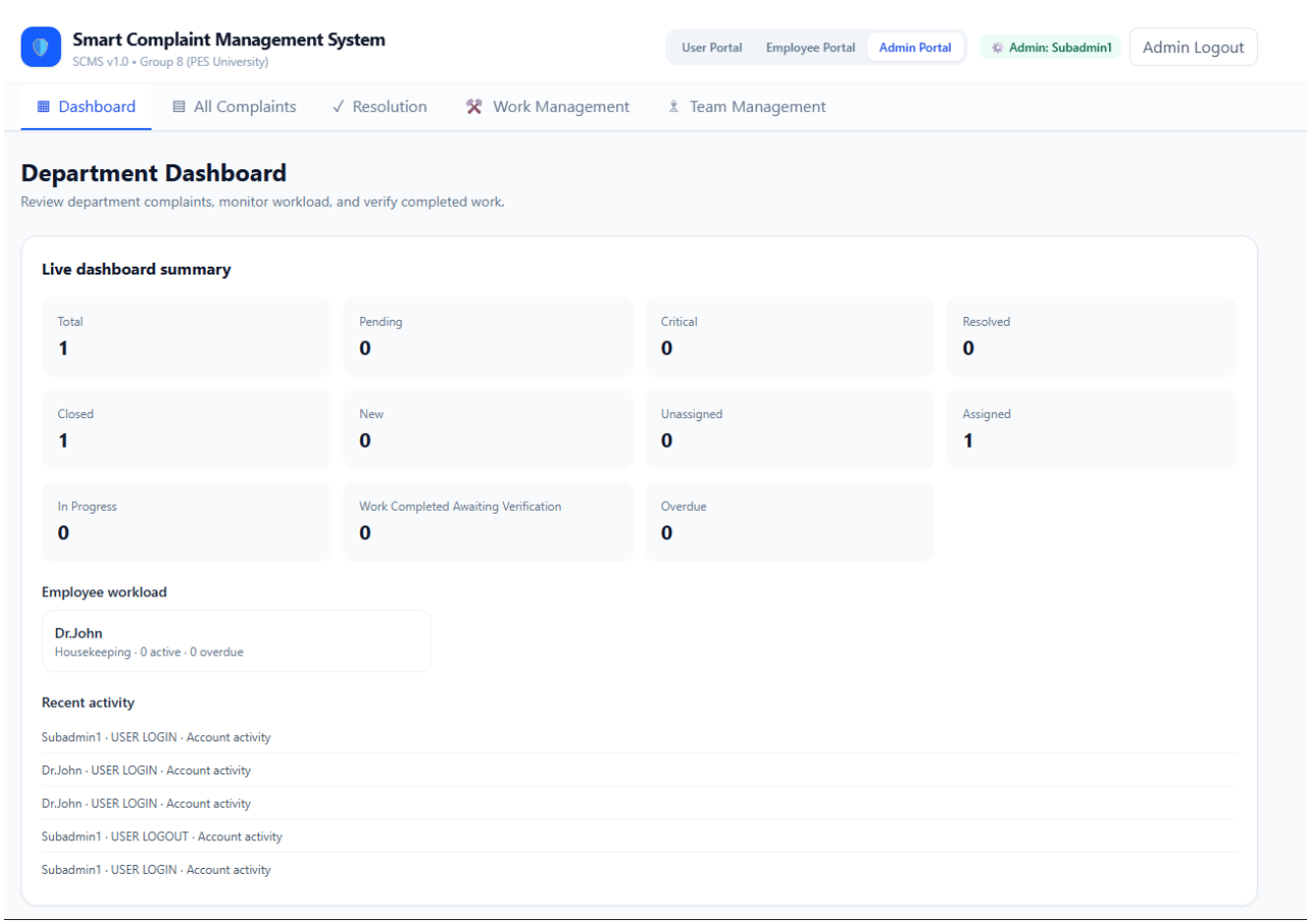
SCMS uses a React/Vite frontend and a Django REST Framework backend. Access is separated across user, employee, department-administration, and super-administration portals. Backend APIs enforce authentication, authorization, complaint lifecycle rules, and data access.

3. Roles and Access

Role	Responsibility / access
User	Submit complaints, view own complaints, track status and updates, see final resolution.
Employee	Access employee portal and assigned work; perform permitted work-progress actions.
Department Admin	Manage complaints and workload within the assigned department; verify completed work as authorized.
Super Admin	Authorized cross-department administration and reopening operations.




Smart Complaint Management System
SCMS v1.0 • Group 8 (PES University)
User Portal
Employee Portal
Admin Portal
Employee: Dr.John
Employee Logout
Dashboard
My Work
✓ Employee authentication successful.
Employee Work Dashboard
Review assignments, record progress, and submit completed work for verification.
My work summary
My Assignments
1
New Assignments
0
Accepted
0
In Progress
0
Completed
1
Overdue
0
Current assignments
CMP-20261109-FD9F837F · Health issue
Housekeeping · LOW · VERIFIED · Due 11/16/2026, 10:33:28 PM
Recent notifications
Complaint closed
CMP-20261109-FD9F837F is now closed.
11/11/2026, 7:53:10 PM
Complaint resolved
CMP-20261109-FD9F837F is now resolved.


Smart Complaint Management System
SCMS v1.0 • Group 8 (PES University)
User Portal
Employee Portal
Admin Portal
Admin: Subadmin1
Admin Logout
Dashboard
All Complaints
Resolution
Work Management
Team Management
Department Dashboard
Review department complaints, monitor workload, and verify completed work.
Live dashboard summary
Total
1
Pending
0
Critical
0
Resolved
0
Closed
1
New
0
Unassigned
0
Assigned
1
In Progress
0
Work Completed Awaiting Verification
0
Overdue
0
Employee workload
Dr.John
Housekeeping · 0 active · 0 overdue
Recent activity
Subadmin1 · USER LOGIN · Account activity
Dr.John · USER LOGIN · Account activity
Dr.John · USER LOGIN · Account activity
Subadmin1 · USER LOGOUT · Account activity
Subadmin1 · USER LOGIN · Account activity

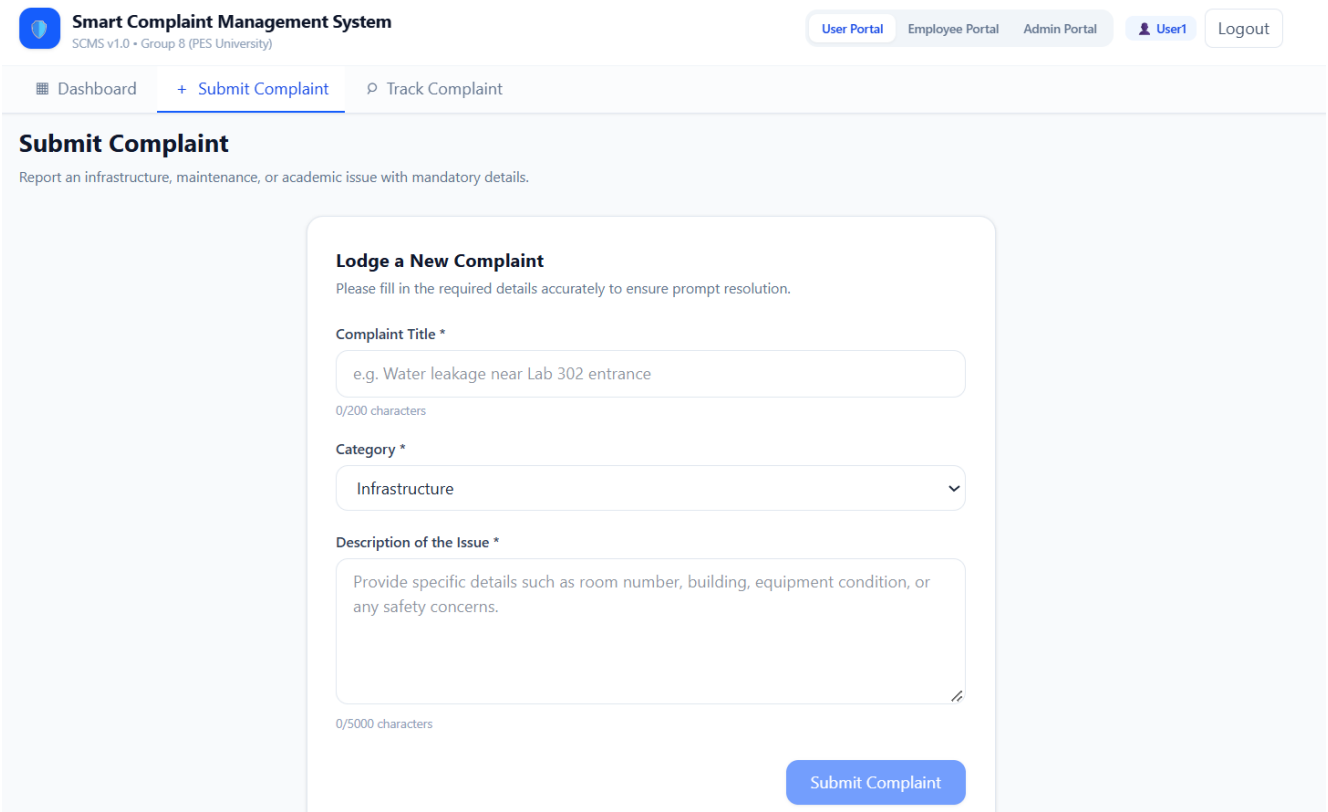
4. Complaint Lifecycle

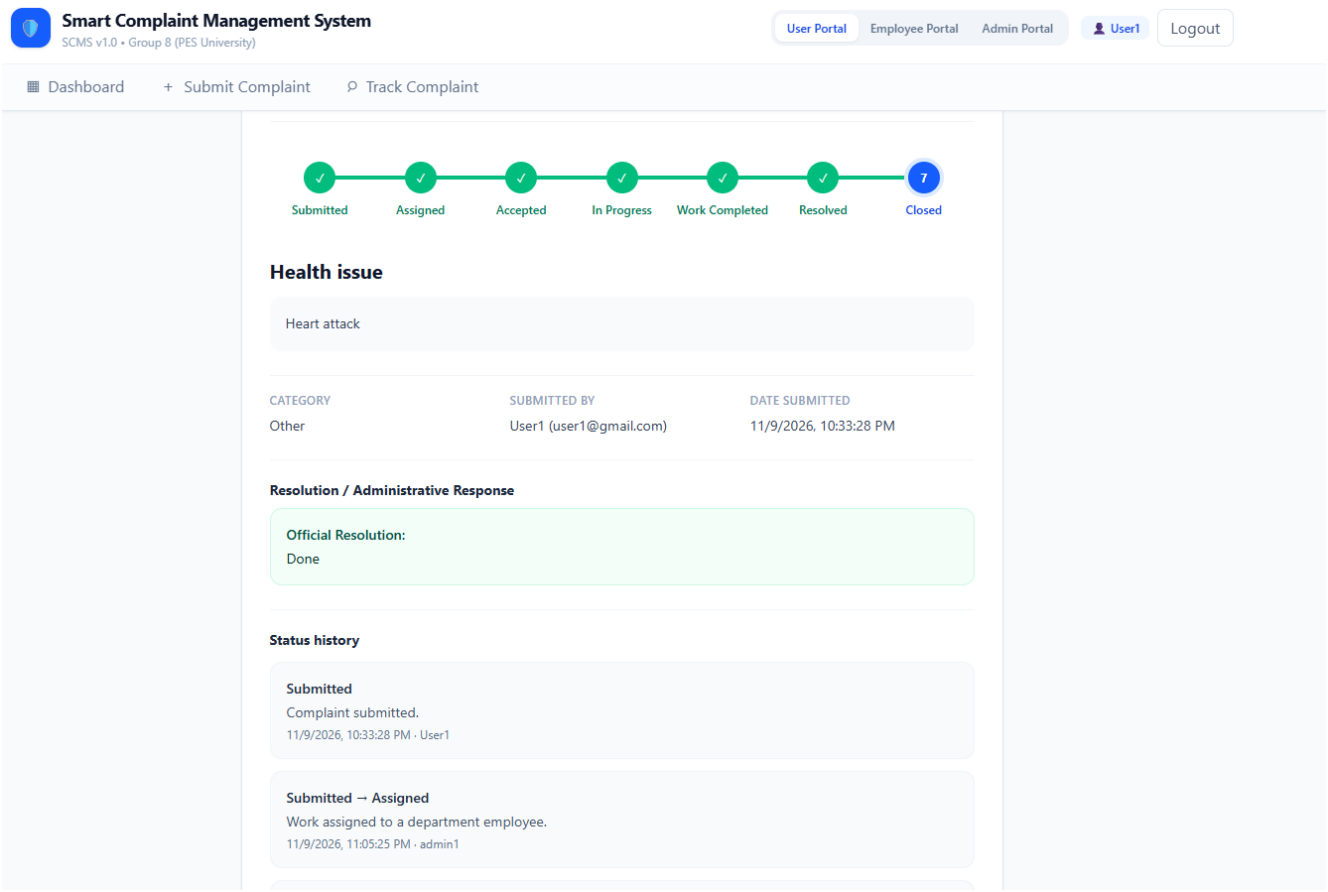
Stage	Meaning
Submitted	Complaint lodged and awaiting processing.
Assigned	A work assignment has been created.
Accepted	Assigned employee has accepted the work.
In Progress	Employee has recorded progress.
Work Completed	Employee has submitted completion evidence / marked work complete.
Resolved	Authorized administrator verified completion and recorded resolution.
Closed	Complaint closed after resolution.

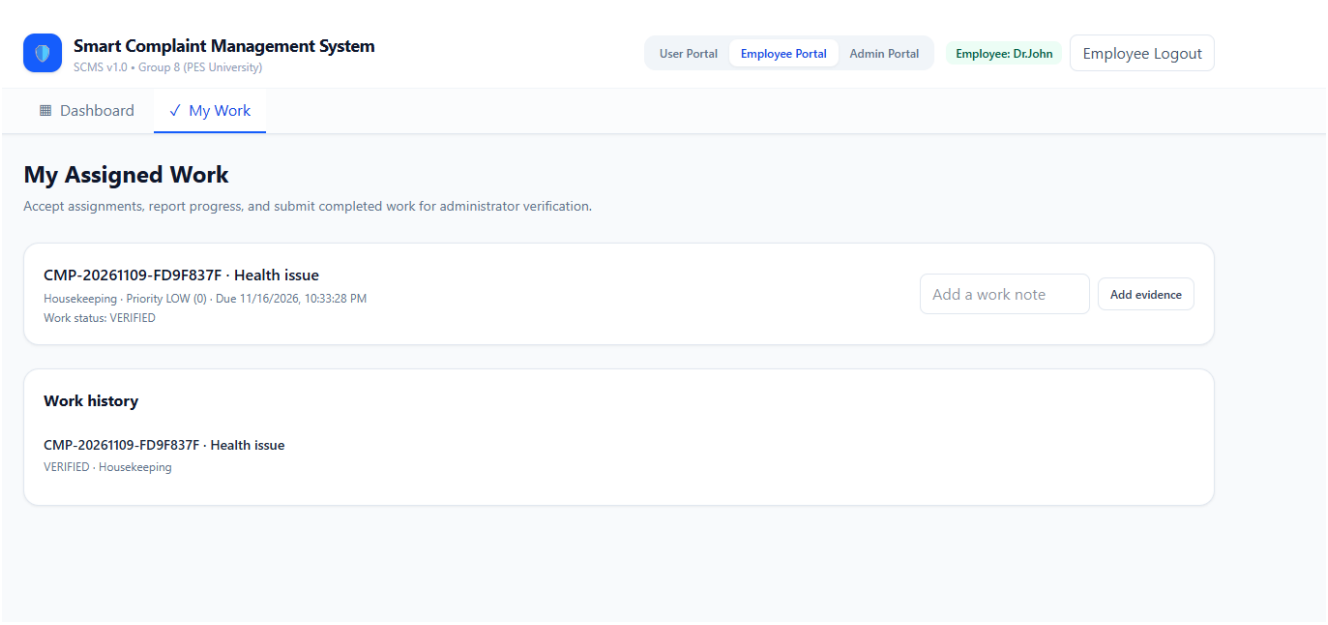
A Super Admin can reopen a closed complaint. Reopening returns it to Submitted, clears active resolution state, retains historical records, and requires a fresh work cycle.

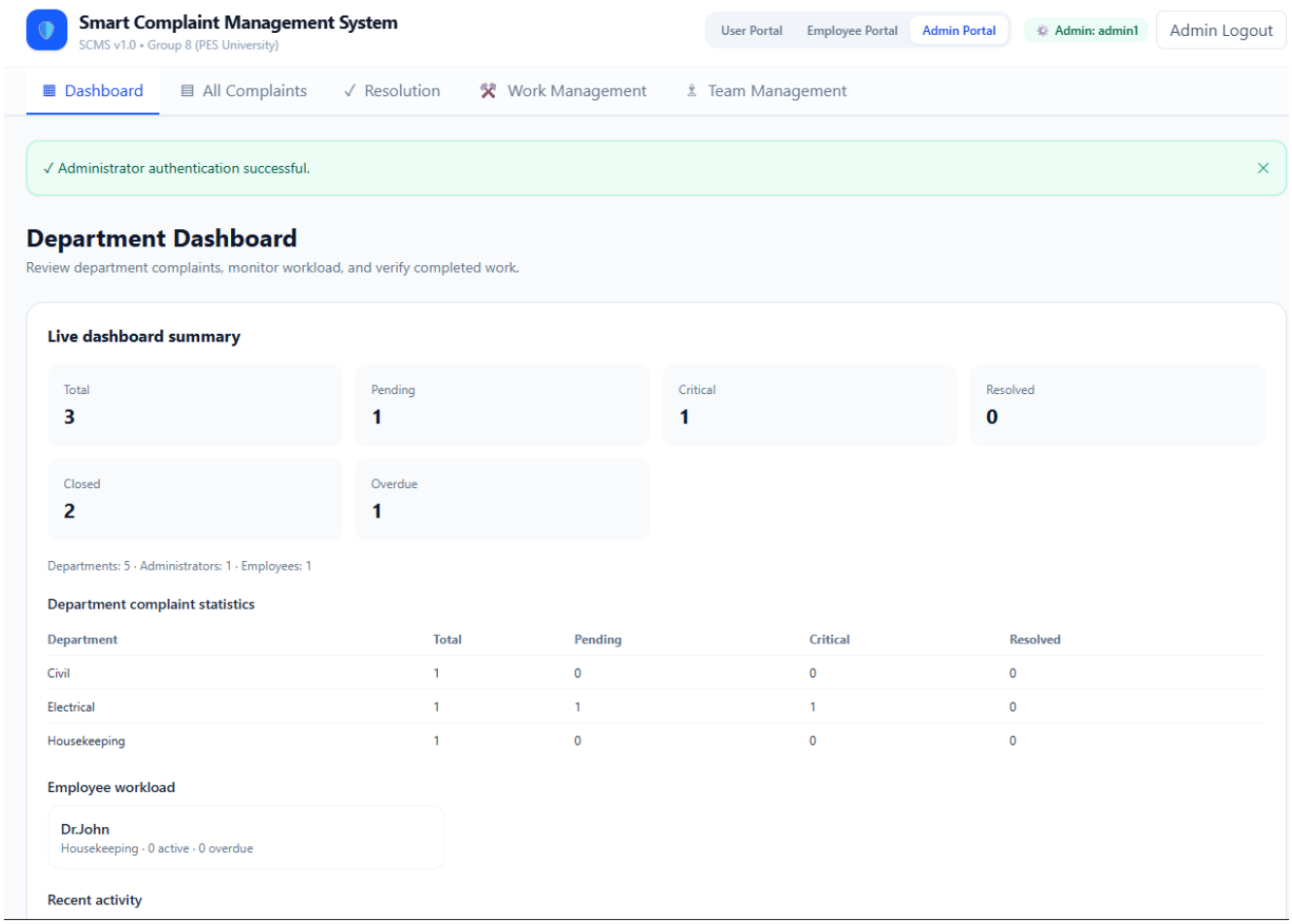
5. Validation Scenarios

Scenario	Validation focus	Result
Portal authentication	Role-specific login and role mismatch rejection	Passed
Complaint ownership	Users access only their own complaint records	Passed
Department isolation	Department Admin visibility remains department-scoped	Passed
Profileless Department Admin	No organization-wide workload exposure	Passed; denied
Employee workflow	Acceptance, progress, completion and evidence authorization	Passed
Administrator verification	Authorized verification before resolution	Passed
Resolution and closure	Resolution persistence and lifecycle restrictions	Passed
Reopening	Permission, active-state reset, history retention	Passed
Audit/history	Important actions recorded and access scoped	Passed
Dashboard	Role-specific data, empty states and scoping	Passed




Smart Complaint Management System
SCMS v1.0 • Group 8 (PES University)
User Portal
Employee Portal
Admin Portal
User1
Logout
Dashboard
Submit Complaint
Track Complaint
Submitted
Assigned
Accepted
In Progress
Work Completed
Resolved
Closed
Health issue
Heart attack
CATEGORY
Other
SUBMITTED BY
User1 (user1@gmail.com)
DATE SUBMITTED
11/9/2026, 10:33:28 PM
Resolution / Administrative Response
Official Resolution:
Done
Status history
Submitted
Complaint submitted.
11/9/2026, 10:33:28 PM · User1
Submitted → Assigned
Work assigned to a department employee.
11/9/2026, 11:05:25 PM · admin1


Smart Complaint Management System
SCMS v1.0 • Group 8 (PES University)
User Portal
Employee Portal
Admin Portal
Employee: Dr.John
Employee Logout
Dashboard
My Work
My Assigned Work
Accept assignments, report progress, and submit completed work for administrator verification.
CMP-20261109-FD9F837F · Health issue
Housekeeping · Priority LOW (0) · Due 11/16/2026, 10:33:28 PM
Work status: VERIFIED
Add a work note
Add evidence
Work history
CMP-20261109-FD9F837F · Health issue
VERIFIED · Housekeeping



6. Security and Data-Access Validation

Portal-specific login endpoints enforce the expected account role before issuing a token.

Regular users are restricted to their own complaint records.

Employee access is limited to employee-specific navigation and permitted work operations.

Department Admin data is scoped to the associated department; a profileless Department Admin is denied workload access.

Super Admin cross-department access is retained for authorized operations.

Evidence and workflow operations include authorization and validation checks.

7. Build and Integration Checks

Check	Result
Django full automated suite	103 tests passed
Frontend lint	0 warnings; 0 errors; 17 files
Frontend production build	Successful; 86 modules transformed
git diff --check	Passed

8. Validation Boundaries

The supplied evidence confirms automated functional/API tests and frontend build checks; it does not establish formal load-test results.

No numeric response-time or throughput benchmark is reported.

SQLite does not provide the same row-level select_for_update locking behavior as PostgreSQL; production concurrency should be validated on the intended deployment database.

Attach actual portal screenshots separately if the evaluator requires visual evidence.

9. Conclusion

Based on the supplied execution results, tested complaint lifecycle behavior, role boundaries, department scoping, assignment workflow, audit/history, dashboard, and frontend build checks passed. Performance and production-database properties not established by the available evidence are explicitly excluded from the conclusions.